

Expressions, Formulae and Brackets — Paper D

HARD • NON-CALCULATOR • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is designed to come out exactly.
- The rules for expanding, factorising and signs are not printed on this paper. You are expected to know them.
- When a bracket is subtracted, the minus sign multiplies every term inside it. This is the single largest source of lost marks in this topic.
- Several questions ask you to explain or to prove something is always true. A calculation on its own is not an explanation — say what the algebra shows.
- The mark for each question is shown in square brackets on the right.
- If a question stops you, leave it and come back — do not spend more than five minutes stuck.

SECTION 1 – EXPANDING AND SUBSTITUTING

1. Expand and simplify:

(a) $6(2x - 3) - 4(x - 5)$ [2]

(b) $3(4x + 2) - 2(3x - 7)$ [2]

2. Given that $p = -2$ and $q = 5$, work out:

(a) $3p^2$ [1]

(b) $(3p)^2$ [1]

(c) $q - p$

[1]

(d) $p^3 + q$

[1]

SECTION 2 – FACTORISING AND REARRANGING

3. Factorise completely:

(a) $24x + 36$

[1]

(b) $14y^2 - 21y$

[1]

(c) $30a + 45b$

[1]

(d) $6x^2 + 9x + 12$

[1]

4. A rectangle has a perimeter of $(14x + 6)$ cm. One of its sides is $(2x + 5)$ cm long.

(a) Write a simplified expression for the sum of two adjacent sides.

[1]

(b) Hence find a simplified expression for the other side.

[2]

- (c)** Check your answer by finding both sides and the perimeter when $x = 3$. [11]

- 5.** Three consecutive even numbers can be written as $2n$, $2n + 2$ and $2n + 4$.

- (a)** Write down the next even number after $2n + 4$. [11]

- (b)** Write a simplified expression for the sum of the three numbers given above. [11]

- (c)** By factorising your answer, show that the sum of three consecutive even numbers is always a multiple of 6. [21]

SECTION 3 – MARKING SOMEBODY ELSE

- 6.** Two students were each asked to expand and simplify $5(2x - 3) - 2(x - 4)$. [51]

Student R wrote: " $10x - 15 - 2x - 8 = 8x - 23$ "

Student S wrote: " $10x - 3 - 2x + 4 = 8x + 1$ "

For each student, describe exactly what they did wrong. Then give the correct answer.

- 7.** A student has written two answers in his book, and both are wrong. [5]
- (i) " $3x + 2 = 5x$, because $3 + 2 = 5$."
- (ii) " $12x + 18$ factorises completely to $2(6x + 9)$."
- For each one, explain the mistake and give the correct answer. For (ii), also state the one-line check that would have caught the error.

End of paper. Check every answer has a unit or a form the question actually asked for.